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Comparative performance of cover systems to prevent acid mine drainage from pre-oxidized tailings: A numerical hydro-geochemical assessment

Thomas Pabst[a,b,*], Bruno Bussière[b,c], Michel Aubertin[a,b], John Molson[d]

[a] Department of Civil, Geological, and Mining Engineering, Polytechnique Montréal, P.O. Box 6079, Succ. Centre-ville, Montréal, QC H3C 3A7, Canada
[b] Research Institute on Mines and Environment, Canada
[c] Department of Applied Sciences, Université du Québec en Abitibi-Témiscamingue, 445 Blvd. de l'Université, Rouyn-Noranda, QC J9X 5E4, Canada
[d] Department of Geology and Geological Engineering, Université Laval, 2325, rue de l'Université, QC G1V 0A6, Canada

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ABSTRACT

The generation of acid mine drainage (AMD) remains a major environmental challenge for the mining industry. The reclamation of old mine sites with pre-oxidized tailings is particularly challenging because of indirect oxidation reactions which can limit the overall effectiveness of an oxygen barrier to prevent AMD. The goal of this project was to quantitatively compare the effectiveness of different cover systems to reclaim two pre-oxidized acid-generating tailings sites, located in Quebec (Canada). Following laboratory column tests, field measurements and observations, coupled hydrogeological and geochemical numerical simulations were conducted to evaluate the effect of various system characteristics. Cover performance was assessed by simulating the evolution of the degree of (water) saturation, pore water pressures, oxygen fluxes and leachate quality. Several reclamation options, including monolayer covers and two- or three-layer covers with capillary barrier effect(s) were simulated. The simulations indicate that because of reduced cover effectiveness with pre-oxidized tailings, the general design targets developed for non-oxidized tailings may not always be directly applicable to already oxidized tailings. The simulations also illustrate how the behaviour and efficiency of a monolayer cover placed over reactive tailings depend on specific factors, including water table position, initial porewater chemistry, and cover materials' hydrogeological properties and thicknesses. The results indicate that under a given set of conditions, a bilayer cover (with a capillary break above the reactive tailings) would not significantly improve cover performance (compared to a monolayer cover) due to water losses by evaporation. The simulations show, however, that a well-designed three-layer cover with capillary barrier effects (CCBE) would be efficient in reducing the oxygen flux and AMD generation, even in the case of highly pre-oxidized tailings. The outcomes from this investigation highlight some of the advantages of carrying out coupled hydrogeological and geochemical simulations to assess the long-term behaviour of reclaimed mining sites.

1. Introduction

Tailings produced by hard rock mines are usually stored in surface disposal facilities often called impoundments. When they contain iron sulfides, tailings can react with oxygen and water to generate acid mine drainage (AMD). AMD is usually characterized by a low pH and high concentrations of dissolved metals and sulfates (e.g. Aubertin et al., 2002; Blowes et al., 2003, 2014; Lindsay et al., 2015; Nordstrom et al., 2015). Effluent treatment can be used during mine operations, but this is neither a sustainable nor a viable solution in the long-term (Aubertin et al., 2015, 2016). It is much more efficient to prevent the oxidation reactions from occurring at the source. Several control and reclamation approaches exist to control oxidation, including various types of cover systems aiming at controlling oxygen and/or water fluxes to the reactive tailings (e.g. Aubertin et al., 2002).

Reclamation of reactive mine waste disposal sites is best achieved when it is planned in advance and integrated into the mining production cycle, according to the principle of "Designing for Closure" (e.g. SRK, 1989; Aubertin and Chapuis, 1991; Hutchison and Ellison, 1992; Aubertin et al., 2002, 2016). This allows preventing the production of acidic waters. However, in some cases, and especially for old mine sites, tailings have been left exposed for decades and can produce large quantities of AMD. The pore water is then already acidic and heavily contaminated. The reclamation of these sites is more complex due in large part to indirect oxidation reactions that occur when the pH is acidic (Nicholson, 1994; Williamson and Rimstidt, 1994; Kirby et al.,

* Corresponding author at: Department of Civil, Geological, and Mining Engineering, Polytechnique Montréal, P.O. Box 6079, Succ. Centre-ville, Montréal, QC H3C 3A7, Canada.
E-mail address: t.pabst@polymtl.ca (T. Pabst).

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1999; Nordstrom, 2000). In this case, Fe(III) remains in solution and can play the role of oxidant. Indirect oxidation reactions are typically much faster than direct oxidation reactions and they produce more acid (e.g. Williamson and Rimstidt, 1994). Indirect oxidation can affect the overall effectiveness of an oxygen barrier to prevent AMD and should be taken into account when designing a cover system for a pre-oxidized tailings impoundment.

The goal of this paper is to quantitatively compare the effectiveness of different cover options to reclaim pre-oxidized tailings impoundments. The paper focuses on the behaviour of two acid-generating tailings disposal sites (sites A and B), located in the Abitibi region, in Quebec (Canada). For site A, the reclamation method initially considered involved the addition of a single layer of slightly alkaline, non-acid-generating tailings (produced at a nearby mine) onto the pre-oxidized reactive tailings, with a limited rise of the water table position in the old impoundment (to help increase the degree of saturation). The tailings impoundment at site B has already been covered by 1 to 2 m of a widely graded and heterogeneous till, extracted in the vicinity of the site; the water table in this case is well below the reactive tailings surface.

The hydrogeological and geochemical behaviour of these two tailings-cover systems was assessed experimentally and numerically to determine the efficiency of monolayer covers under the actual or anticipated field conditions. The program included fieldwork and a series of column tests conducted in the laboratory (Pabst, 2011; Pabst et al., 2014, 2017a,b). The experimental data obtained from the laboratory tests were used to validate and calibrate numerical models constructed with Vadose/W (GeoSlope Int. Ltd., version 2007) and MIN3P (Mayer and Benner, 1999; Mayer et al., 2002). The effect of various influence factors, such as climatic conditions, cover thickness, depth of the water table and material properties, were specifically addressed (Pabst, 2011; Pabst et al., 2017a,b).

In this paper, alternative cover configurations are analysed, using the same cover materials and other materials available locally. The efficiency of the cover systems at limiting oxygen ingress is evaluated using different approaches. First, the oxygen flux reaching the reactive tailings is evaluated for each system and set of conditions, using the code Vadose/W (Geo-Slope Int. Ltd.). A cover is usually considered efficient when it can achieve at least the same performance as a 1 m thick (still) water cover, with an oxygen flux equal to or less than about 50 $g/m^2/y$ (Aubertin et al., 1999; Dagenais, 2005). The degree of saturation (S_r) in the moisture retaining layer (MRL) is also a good indicator for the efficiency of an oxygen barrier; a degree of saturation $S_r > 85\%$ usually implies that oxygen diffusion will be low enough to prevent AMD (e.g. Ricard et al., 1997, 1999; Bussière et al., 2003; Mbonimpa et al., 2003; Dagenais et al., 2006). Finally, the simulated water quality obtained from reactive transport modelling carried out with MIN3P (Mayer and Benner, 1999; Mayer et al., 2002) is also used as a direct indication of the cover performance. The combination of these various indicators, obtained using two different numerical codes, helped assess the performance of these cover systems.

This paper presents and discusses the main results from this parametric analysis, which complements previous experimental work, and provides recommendations for the reclamation of pre-oxidized tailings impoundments in relatively humid climates. Previous work mainly focused on monolayer covers coupled with an elevated water table (Pabst et al., 2017a,b). This paper compares the performance of monolayer, bilayer and multilayer covers to control AMD generation from pre-oxidized tailings. This assessment is based on coupled hydrogeochemical simulations, calibrated on experimental results. The effect of climatic conditions, water table depth, tailings and cover material properties, and cover thickness are specifically addressed.

The approach and results presented here have helped the decision process for selecting the final cover design at one of the targeted tailings disposal sites.

2. Background on reclamation of acid-generating tailings sites

Under relatively humid climatic conditions, i.e. where the water balance is positive most of the year, prevention of AMD usually relies on controlling oxygen flux with water or a nearly saturated porous medium (e.g. Collin, 1987; Nicholson et al., 1989; Aachib et al., 1994, 1998, 2004). Water covers typically consist of disposing reactive mine tailings under at least 1 m of water (e.g. Li, 1997; Yanful and Catalan, 2002; Mbonimpa et al., 2008; Awoh et al., 2013). The oxygen diffusive flux in water being approximately 10,000 times lower than in air, the oxygen supply is then sufficiently small that the generation of AMD remains negligible. Although generally effective for preventing the oxidation of reactive tailings, this approach can be complex and costly to implement, particularly for older artificial impoundments. Also, building dikes that will last the (indefinite) lifetime of tailings ponds is a major challenge (Aubertin et al., 1997, 2016; Vick, 2001).

Another approach to control AMD generation relies on the use of engineered soil covers. These include multilayered covers made of several soils placed on top of each other (Aubertin and Chapuis, 1991; Hutchison and Ellison, 1992; Aubertin et al., 2002). The number of layers can vary depending on the design, the climate and the position of the water table (Aubertin et al., 1995, 1999). For instance, a typical 3-layer-cover is made of a 0.5 to 1 m thick MRL made of fine-grained material, sandwiched between two 30 to 50 cm thick coarse-grained material layers (Aubertin et al., 1994, 1999, 2006; Bussière et al., 2003, 2006; Bussière, 2007). The coarser material layers are expected to drain easily and rapidly, reaching a low volumetric water content and low unsaturated hydraulic conductivity, thus preventing both water rise and loss by evaporation and drainage from the fine-grained layer. The MRL can therefore remain nearly saturated over a relatively long time, hence limiting oxygen diffusion to the underlying reactive tailings. Over time, this fine-grained material could eventually desaturate, but it can be expected that natural precipitation would provide sufficient recharge so it can remain highly saturated at all time. These types of covers, which are referred to as covers with capillary barrier effects (CCBEs), have been proven efficient at several mine sites, including Les Terrains Aurifères - LTA (McMullen et al., 1997; Ricard et al., 1999; Bussière et al., 2006), and Lorraine (Dagenais et al., 2002, 2005).

As an alternative to water covers and multilayered covers, the elevated water table (EWT) approach relies on controlling the water table level and degree of saturation of the reactive tailings (Orava et al., 1997). The technique essentially consists of raising and maintaining the water table to a depth less than the height of the capillary fringe (Dagenais, 2005; Ouangrawa et al., 2006, 2009, 2010). A layer of granular material (a monolayer cover) is usually placed above the reactive tailings with an EWT. A coarse-grained material cover helps with the recharge and limits water loss by evaporation (Dagenais et al., 2006), while a fine-grained material cover may promote water retention (through the capillary rise) and prevent oxygen diffusion (Demers et al., 2008). The efficiency of the monolayer cover is then highly dependent on the water table position (e.g. Cosset and Aubertin, 2010; Dobchuk et al., 2013). When well designed and implemented, the EWT technique can be quite efficient to limit the generation of AMD (Dagenais, 2005; Ouangrawa, 2007; Demers et al., 2008, 2009; Ethier et al., 2014, 2016).

While having been fairly well studied, the performance of mono- and multilayered cover systems has, until now, mainly been assessed for non-oxidized tailings, and the design recommendations (e.g. layer thickness and properties, water table position) made only for fresh tailings (at ongoing or recently closed mining sites). One of the objectives of this paper is therefore to assess cover performance for pre-oxidized tailings at two mine sites and to compare the results with general design targets developed for non-oxidized tailings.

3. Methodology

3.1. Materials

Four types of materials were considered in the present study: reactive tailings TA and TB, non-reactive tailings CA, and till cover material CB. Reactive tailings TA and TB were sampled from sites A and B, respectively. Non-acid-generating tailings, CA, used as cover material on tailings TA, were provided by the operating mine involved in the reclamation work. The till CB used for the cover on site B, was sampled at the same time as tailings TB. These four materials have been extensively characterized in the laboratory (Pabst, 2011; Pabst et al., 2014), while field data, including in-situ porosity, reaction rate coefficient, volumetric water content and water table depth, were also measured (Pabst, 2011).

The water retention curve (WRC) was also evaluated for each material, using modified Tempe cells, with the experimental results fitted with the van Genuchten (1980) model. The saturated hydraulic conductivity k_{sat} was measured in flexible wall permeameters. The experimental k_{sat} values obtained from these permeameter tests were compared (and validated) with predictive estimates obtained from the Kozeny-Carman (KC) equation (Chapuis and Aubertin, 2003) and the Kozeny-Carman modified (KCM) equation (Mbonimpa et al., 2002). The permeability function was obtained using the Mualem (1976) formulation based on the approximate analytical solution proposed by van Genuchten (1980); the unsaturated hydraulic conductivity of the different materials under various degrees of saturation could thus be estimated. The parameters used in the numerical models are summarized in Fig. 1 and Table 1.

In addition to the two reactive tailings TA and TB, and the two cover materials CA (non-acid generating tailings) and CB (natural till), two non-acid-generating waste rocks (WR1 and WR2) were also considered in this study. Their coarse grain size distribution can be expected to contribute to the development of a capillary break with the finer materials CA and CB in two- or three-layer CCBEs. The properties of the waste rock WR1 and WR2 were taken from Dawood et al. (2011); (Fig. 1 and Table 1). Both waste rocks have a hydraulic conductivity comparable to a sandy soil. WR2 is slightly coarser than WR1, which has little effect on the saturated hydraulic conductivity but is more noticeable with the estimated water retention curves (Fig. 1). The residual water content θ_r is relatively high (around 0.06, i.e. $S_r = 24\%$) because of the significant proportion of fine-grained particles in these waste rocks (Dawood et al., 2011; also see Peregoedova, 2012).

Oxygen consumption tests conducted for this project (Gosselin, 2007; Gosselin et al., 2012) gave a reaction rate coefficient $K_r = 3 \times 10^{-5}\,s^{-1}$ for tailings TA, and $K_r = 1 \times 10^{-5}\,s^{-1}$ for tailings TB. The cover materials CA and CB can be considered non-reactive, as no significant oxygen consumption was measured. Waste rocks WR1 and WR2 are also assumed to be non-reactive. The effective diffusion coefficient D_e was measured in diffusion cells following the procedures described by Mbonimpa et al. (2003, 2011) and Aachib et al. (2004). Experimental results indicate that the Collin and Rasmuson (1988) model (used in Vadose/W) and the related Aachib et al. (2004) model (used in MIN3P), accurately represent the effective diffusion coefficient D_e (Pabst et al., 2014).

Table 1
Physical properties and values of the van Genuchten (1980) model parameters for the materials considered in the simulations. The saturated hydraulic conductivity k_{sat} is also given. θ_s: saturated water content (or porosity); $\theta_{r,vG}$: fitted residual water content; α_{vG} and n_{vG}: van Genuchten equation parameters.

	Classification[a]	θ_s (−)	$\theta_{r,vG}$ (−)	α_{vG} (m^{-1})	n_{vG} (−)	k_{sat} (m/s)
TA	ML - Silt	0.40	0.0	0.042	1.35	4×10^{-7}
CA	ML - Sandy silt	0.47	0.0	0.21	2.10	6×10^{-7}
TB	ML - Silt with sand	0.36	0.0	0.30	1.40	3×10^{-7}
CB	SM - Silty sand	0.36	0.0	0.65	1.70	3×10^{-6}
WR1	GM - Silty gravel with sand	0.25	0.06	3.3	1.89	5×10^{-5}
WR2	GM - Silty gravel with sand	0.25	0.06	14.5	3.50	1×10^{-4}

[a] Based on the Unified Soil Classification System (ASTM D2487, 2011).

The mineralogical compositions of reactive tailings TA and TB and cover materials CA and CB are given in Table 2; these values were included in the hydrogeochemical model MIN3P.

Reactive tailings TA and TB contain a relatively high proportion of pyrite (respectively around 12.0% and 4.0% by weight). Cover material CA contains about 5.5% calcite and 2.0% dolomite, while 2.0% calcite (but no dolomite) was measured in cover material CB. Both CA and CB are rich in albite (respectively 47.0% and 28.5% by weight). A relatively high content of anorthite (10%) was identified in the cover material CA. XRD measurements were repeated on each of these four materials before and after the column tests but no significant variations were observed. Waste rocks WR1 and WR2 (not used in these column tests) were assumed non-reactive (no sulfides) and, for the purpose of the models, assumed made only of quartz.

3.2. Laboratory column tests

Large instrumented columns (230 cm high, 14.5 cm internal diameter) were set up in the laboratory to provide representative large-scale experimental results for the validation and calibration (if needed) of numerical models. Columns contained 170 cm of reactive tailings (TA or TB) covered by 30 to 40 cm of cover materials (CA or CB). A total of 19 wetting and drying cycles were applied during the column tests.

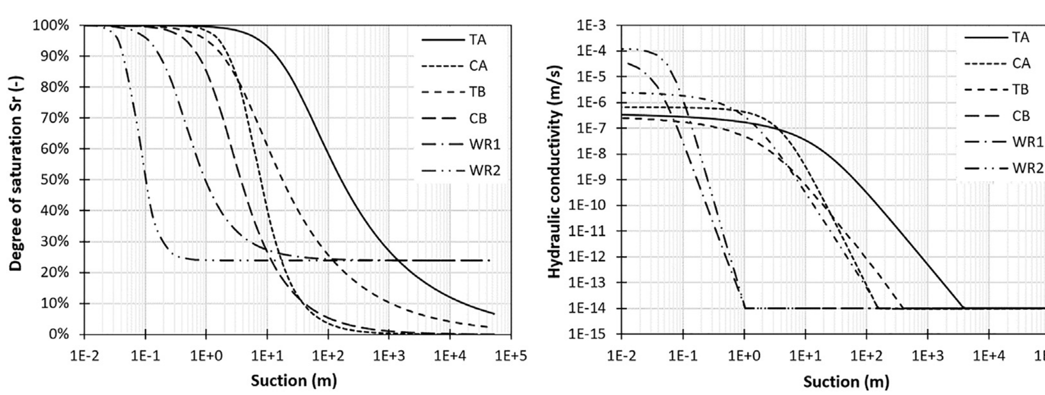

Fig. 1. Water retention curves and unsaturated hydraulic conductivity functions used in the simulations. TA and TB are the reactive tailings to be covered. CA is a non-acid-generating tailings and CB a till; both have been used for monolayer covers and for the MRL in multilayer covers. WR1 and WR2 are two waste rocks available in the vicinity of mine site A which were considered for the construction of the CCBEs.

Table 2
Mineralogical compositions of the four materials used in the column tests (wt% in $m_{mineral}^3\ m_{solids}^{-3} \times 100\%$); rounded mean values based on five XRD measurements.

	TA	TB	CA	CB
	wt%	wt%	wt%	wt%
Pyrite	12.0	4.0	n.d.	n.d.
Quartz	55.0	67.5	21.5	48.5
Calcite	n.d.	n.d.	5.5	2.0
Dolomite	n.d.	3.0	2.0	n.d.
Chlorite	5.0	6.5	6.6	2.0
Muscovite	22.0	15.0	7.4	13.0
Albite	1.0	2.0	47.0	28.5
Anorthite	5.0	1.0	10.0	6.0
Gypsum	n.d.	1.0	n.d.	n.d.

n.d.: not detected.

The reference position of the water table was set at 90 cm below the base of the columns, i.e. about 260 cm below the reactive tailings surface. The top of each column was open to the atmosphere. Each column was instrumented with pore water pressure, water content and oxygen sensors. Water was regularly added at the top of the columns and sampled at their base and analysed. A detailed description of the column tests can be found in Pabst (2011) and Pabst et al. (2014). The methodology is similar to that developed over several years by the authors' research group (Aachib et al., 1994, 1998; Aubertin et al., 1995, 1999; Bussière et al., 2004; Ouangrawa et al., 2010; Demers et al., 2011).

3.3. Validation of numerical simulations

Vadose/W 2007 (GeoSlope Int. Ltd.) was used in this study to simulate water flow and oxygen diffusion in the tailings-cover systems. Vadose/W is a finite element code that includes variable climatic boundary conditions, surface ponding, and soil-atmosphere liquid and gaseous phase exchange. This commercial code has been commonly used to assess the behaviour of tailings and covers (e.g. Shackelford and Benson, 2006; Adu-Wusu and Yanful, 2007; Gosselin, 2007; Cissokho, 2007; Demers et al., 2009; Martin et al., 2010; Kalonji Kabambi et al., 2017). Oxygen flux in Vadose/W is simulated using reaction rate coefficients reported above. Evaporation in Vadose/W is simulated using the Penman-Wilson equation (Wilson, 1990).

Reactive transport (hydro-geochemical) simulations were conducted using MIN3P, a 3D-finite volume numerical model developed for simulating 3D reactive transport, including variably-saturated flow, diffusive gas transport, sulfide mineral oxidation, and kinetic or equilibrium-controlled geochemical reactions (Mayer and Benner, 1999; Mayer et al., 2002). The model has been previously tested, verified and applied to several studies of acid mine drainage (e.g. Jurjovec et al., 2004; Molson et al., 2005, 2008; Ouangrawa et al., 2009; Demers et al., 2013).

Sulfide oxidation reactions and oxygen fluxes were simulated in MIN3P based in part on the input mineralogy (Table 2). Although only direct oxidation is simulated in MIN3P, indirect oxidation reactions were taken into account by calibrating the reactive transport model based on column tests (Pabst et al., 2017b). Since MIN3P cannot simulate evaporation the same way as Vadose/W does, water fluxes (either positive or negative) at the surface of the simulated columns were therefore imported from the Vadose/W simulation results and applied as boundary conditions in the MIN3P column models. The resulting water content and suction profiles within the tailings and cover systems were identical to Vadose/W results. Also, oxygen fluxes simulated in Vadose/W were input as boundary conditions in MIN3P to facilitate the comparison between the two codes.

3.4. Simulation of basic field conditions

Numerical models were validated and calibrated using the results from laboratory column tests. A detailed description of the numerical modelling of the column tests and their calibration is presented by Pabst et al. (2017a) for the hydrogeological behaviour simulations and Pabst et al. (2017b) for the hydrogeochemical (reactive transport) response models. After the numerical models were validated, they were expanded to simulate alternate scenarios, initially based on field conditions observed at the two mine sites (Abitibi region, Quebec). A climatic boundary condition, based on daily local field data (including air temperatures, relative humidity, precipitation and wind speed) obtained from Environment Canada (Fig. 2), was applied at the surface of the model. Year 2009 was used as a reference case (most of the fieldwork was carried out that year). The simulations were run for ten years with the same climatic boundary conditions each year. Hydrogeological simulation results showed no significant variations from one year to the next and are thus only presented over a single year (year 10). During the four winter months (Dec. – March), the surface of the simulated site was considered frozen, and consequently there was no water or gas movement in the cover during this period.

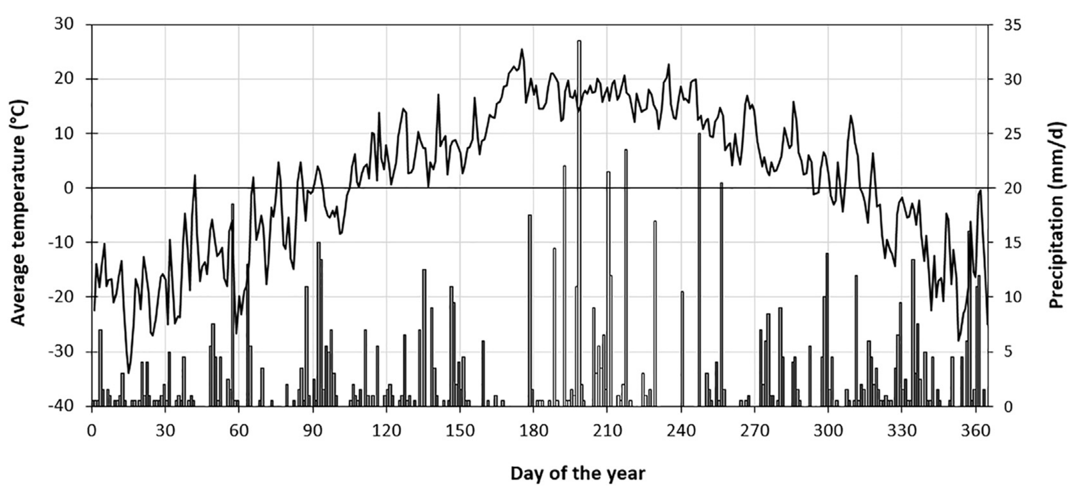

Fig. 2. Average daily temperatures (°C; continuous line) and precipitation (mm/d; histogram) over one year (Environment Canada, 2009). Day 1 corresponds to January 1st. Winter usually lasts from early January to the end of March (approximately day 1 to day 90), when melting of snow and ice lenses occurs. The warmest period of the year is July and August (day 180 to day 240). After the autumn, winter starts again around day 330 (December). Some simulated cases include a dry period (without precipitation or recharge in July and August), indicated with open bars.

In some cases, the environmental conditions applied in the model were altered to simulate a drought and to assess the corresponding response of the cover system(s). Dry periods can significantly affect the performance of oxygen barriers that rely on precipitation (recharge) to replenish water losses and maintain a high degree of saturation. For these simulations, the imposed recharge in the months of July and August (day 180 to day 240) was nil, with the other parameters remaining unchanged; in such cases, the yearly precipitation was reduced to 631 mm, while it is around 844 mm for a normal year. A two-month drought in this part of Canada is rather extreme, and has been considered a worst-case scenario for cover design (e.g. Ricard et al., 1997, 1999).

Different cases with various water table elevations were simulated (water table elevation in-situ may vary over time and location); the considered design in the various simulations was −2 m, −4 m and −6 m below the tailings-cover interface. An additional case with a water table located 1 m below reactive tailings TA was also simulated (such a situation is not representative of current or achievable conditions for site B). The water table was imposed at the base of the models (10 m below the reactive tailings surface) to avoid numerical conflicts between the climatic and hydrogeological boundary conditions. The simulated water table during severe evaporative phases can therefore be deeper than imposed. This allows for a more accurate assessment of the EWT technique and the performance of the monolayer cover as an evaporation barrier.

3.5. Simulated cover configurations

Three different types and configurations of oxygen barrier cover systems were simulated (Fig. 3):

- monolayer covers made of non-acid generating tailings CA or till CB, with different thicknesses varying from 1 to 4 m;
- bilayer covers with a 1 m to 3 m thick MRL made of CA or CB installed on top of 1 m of coarse-grained material (WR1 or WR2) which acts as a capillary break layer;
- multilayer covers, where a 1 m thick retention layer (CA or CB) is placed between two coarse-grained material layers (WR1 or WR2); the coarse layer at the base is 1 m thick, and the coarse layer on top is 0.5 m. This layered system represents a CCBE designed to act as an oxygen barrier (e.g. Aubertin et al., 1995, 1999, 2002; Bussière et al., 1998, 2003).

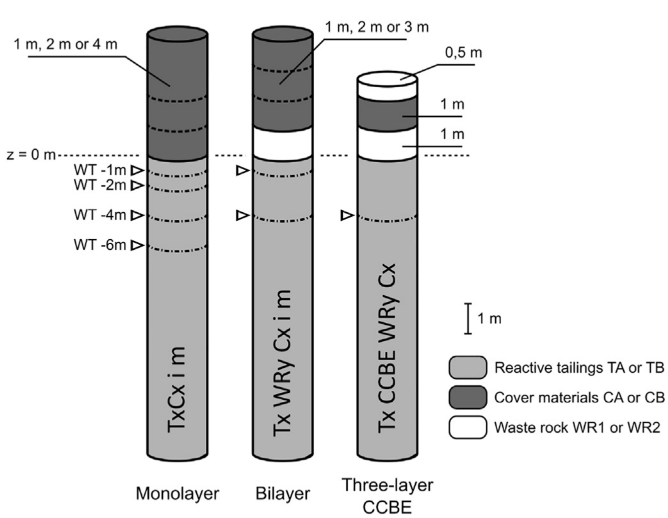

Fig. 3. Conceptual layout of the simulated cover systems (see text for details). WT: water table; x: A or B; y: 1 or 2; i: thickness of Cx layer, (between 1 and 4 m); z = 0 m is the tailings-cover interface.

The results of the hydrogeological simulations are first presented in terms of degree of saturation and pore water pressures in critical profiles. Only the top 5 m of the tailings vertical profile are shown in the figures for clarity (reactive tailings are actually 10 m thick in the models). Annual oxygen fluxes at the surface of the cover are also calculated as cumulated daily values over one year (sum of oxygen fluxes calculated for each day of the simulations, including a zero flux over the winter months).

4. Results

4.1. Monolayer covers for different water table positions

4.1.1. Hydrogeological behaviour

The effect of the water table depth on the performance of 1 m thick monolayer covers was investigated for non-acid-generating tailings CA and till CB. The water table depth was varied from 1 m (for TACA) or 2 m (for TBCB) to 6 m below the tailings-cover interface.

The simulated degree of saturation is shown over time at a position 10 cm above the tailings-cover interface (Fig. 4a). Results are shown for different water table depths, over a one-year period starting on January 1st (day 1). The evolution of the degree of saturation in the cover follows several stages. First, during the winter months at the beginning of each year (day 1 to 90 approximately), the cover is frozen, and it is assumed that there is no oxygen flux through the cover. Upon thawing in April (day 90), the degree of saturation in the cover rapidly increases due to snowmelt and infiltration, and reaches its maximum value within a few days or weeks. The cover remains at that degree of saturation for up to about 20 days, depending on the cases. The degree of saturation then progressively decreases due to both downward drainage and evaporation. This decrease lasts for about 50 to 80 days until mid-summer (day 200). The degree of saturation then oscillates for the rest of the year, as it varies with water losses and rain events during autumn. Frost starts again in December (day 330).

The degree of saturation in the cover depends on the water table position as it tends to decrease when the water table depth increases. For a 1 m thick monolayer cover, the minimum value of the degree of saturation (10 cm above the tailings-cover interface) around mid-summer (day 200) for cover CA is about 50% when the water table is 6 m below the tailings-cover interface, 60% for a depth of 4 m, and around 80% and 90% for a depth of 2 m and 1 m, respectively.

The degree of saturation in the monolayer cover is also a function of the cover material properties. The degree of saturation for cover CB (sandy till) is typically lower than for CA (silty tailings). For example, around day 200, S_r is between 10 and 20% lower in CB than in CA (independently of the water table position).

The vertical profiles of the degree of saturation in mid-summer (around day 200, which corresponds to one of the lowest values of S_r; Fig. 4b) indicate that covers are dryer closer to the surface. The difference in degree of saturation between the top and the bottom of the cover can vary from only a few percent to > 40%, depending on the water table position and cover material. Results also show that reactive tailings TA and TB are close to saturation (S_r > 95%) at that point, even when the water table is as deep as 6 m below the tailings-cover interface.

The pore water pressure profiles (Fig. 4c) indicate that the effect of evaporation is significant, creating high suctions close to the surface. This suction produces an upward flow component that contributes to moisture loss to the atmosphere. Evaporation also has a major influence on the pore pressures throughout the cover, but it may have a more limited effect on the underlying reactive tailings, depending on the cover material. With cover material CA, the suction in tailings TA is much higher than at equilibrium. For example, the pore water pressure head at a depth of 1 m below the tailings-cover interface (WT −1 m) is around −1.1 m, −2.1 m at WT −2 m, −4.1 m at WT −4 m and −5.9 m at WT −6 m, while it should be, in the absence of evaporation,

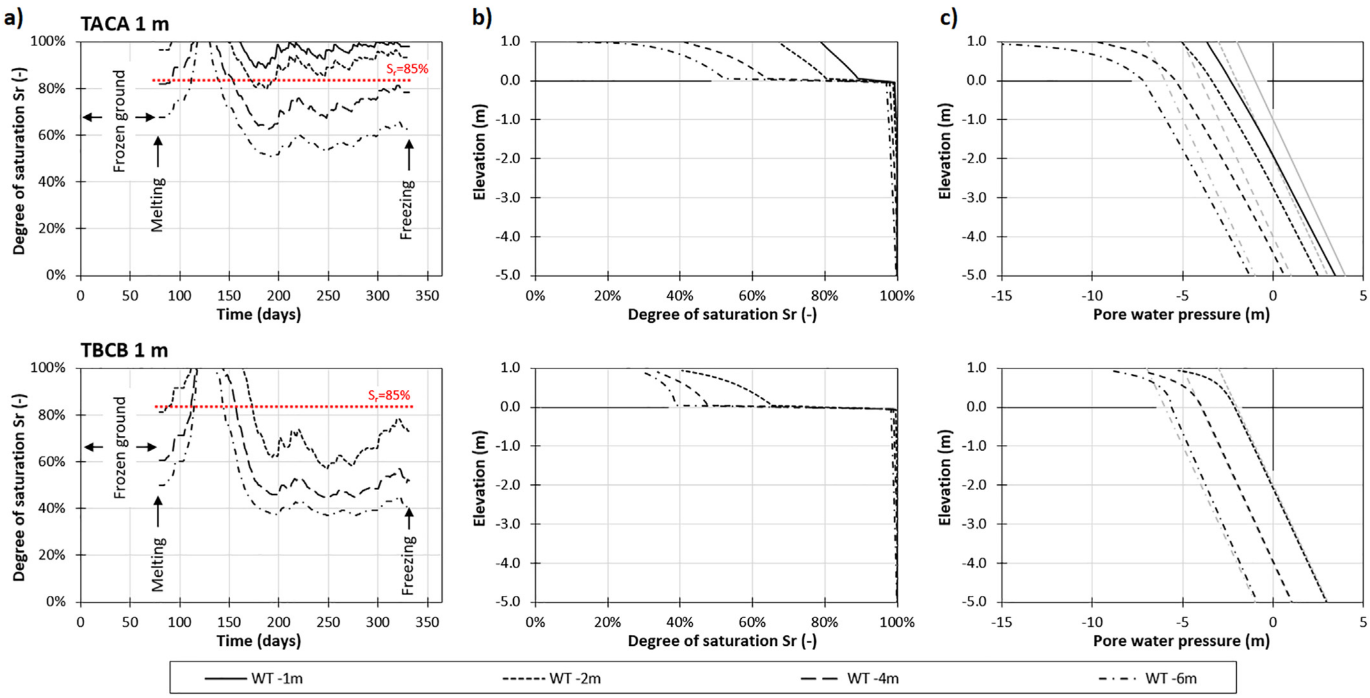

Fig. 4. Hydrogeological behaviour of a 1 m thick monolayer cover (materials CA or CB) placed over acid-generating tailings (TA or TB) for various water table positions. (a) Evolution of the degree of saturation in the cover, 10 cm above the surface of the reactive tailings (z = 0.1 m) over one year; (b) vertical profiles of the degree of saturation and (c) pore water pressure at day 200 (mid-summer). Pale grey lines in (c) indicate pore water pressure profiles at equilibrium in the absence of climatic conditions (hydrostatic conditions).

0 m, −1 m, −3 m and −5 m, respectively. These values indicate that the 1 m thick monolayer cover CA (which has hydrogeological properties similar to tailings TA; Fig. 1) does not prevent evaporation (and moisture loss) from the underlying reactive tailings and that the positive effect of an EWT cannot be maintained under these conditions. This explains why tailings TA are somewhat desaturated close to the surface, even though their air entry value (AEV ≈ 6–8 m) is higher than the depth to the water table. In this case, evaporation affects reactive tailings TA over a depth of several meters. With cover material CB, however, the difference between the expected and simulated suction 1 m below the tailings-cover interface is only a few centimetres. These results confirm that cover material CB (sandy till) is more effective to prevent evaporation from the underlying tailings than CA (silty tailings), because of a stronger hydrogeological contrast between CB and TB than between CA and TA. These results are in agreement with those of Dagenais et al. (2006). In the TBCB case, evaporation affects only the top 70 cm or so of the monolayer cover; in this instance, desaturation of tailings TB is essentially due to downward drainage and to their relatively low AEV (≈2 m).

4.1.2. Oxygen fluxes

Calculating the oxygen flux at the surface, through a cover system, can be performed to assess its efficiency to limit oxidation of the reactive tailings. The oxygen flux through a cover varies over time depending on the local and global degree of saturation (Fig. 5), while oxygen flux is zero during the winter when the ground is frozen. It remains very low upon snow melting, while the cover remains nearly saturated (until around day 130; Fig. 4). Oxygen flux then increases quickly when covers CA and CB desaturate; the dryer the cover, the greater the increase in oxygen flux. The flux then varies during the summer and autumn, increasing during dry spells, and decreasing after rain events. Daily oxygen flux with CA can reach 10 g/m^2/day when the water table is located 6 m below the surface of tailings TA, but remains below 1 g/m^2/day when it is 1 m deep. Daily oxygen fluxes with CB are between 3 and 5 g/m^2/day during the second half of the year, and increase when the water table is deeper.

Cumulative annual fluxes (sum of daily values over one year) for the different configurations of a monolayer cover, are summarized in Table 3. The simulated fluxes confirm the previous observations that a monolayer cover CA can be efficient (i.e. oxygen flux ≤50 g/m^2/y) when the water table is located < 2 m below the reactive tailings-cover interface. A water table located 1 m or less below the interface can be very efficient, with simulated oxygen fluxes < 7 g/m^2/y, even with a relatively thin (1 m) cover. For tailings TB, the relatively coarse grain size of the till leads to desaturation of the cover CB. The oxygen flux is then > 50 g/m^2/y, even though tailings TB are less reactive than TA. Additional calculations (not shown here; see Pabst, 2011) indicate that these two single layer covers are even less efficient during dry years, due to more pronounced desaturation of covers CA and CB.

4.1.3. Water chemistry

The assessment of the oxygen flux only partially accounts for the contribution of indirect oxidation (which does not require oxygen) to the generation of AMD. Reactive transport simulations were therefore carried out to more accurately assess the effectiveness of the cover systems to prevent AMD generation from the already partially oxidized tailings TA and TB. These simulations were carried out over 10 years to properly assess the trends that may develop after installation of the cover.

The initial pore water chemistry assumed in the simulations was based on laboratory column tests (Pabst et al., 2017b) and the initial pH can therefore be relatively high, as was observed in the column tests. Samples collected in situ contained buffering minerals (Table 2), but these were removed in the models (and replaced by quartz) to allow for a more direct comparison of the performance of the various cover systems in terms of AMD generation. In the longer term, this also matches in situ observations showing that pore waters at sites A and B are already strongly acidic (e.g. Aubertin et al., 1995, 1999; Pabst, 2011; Ethier et al., 2014, 2016), indicating that the buffer capacity has already been exhausted at these locations.

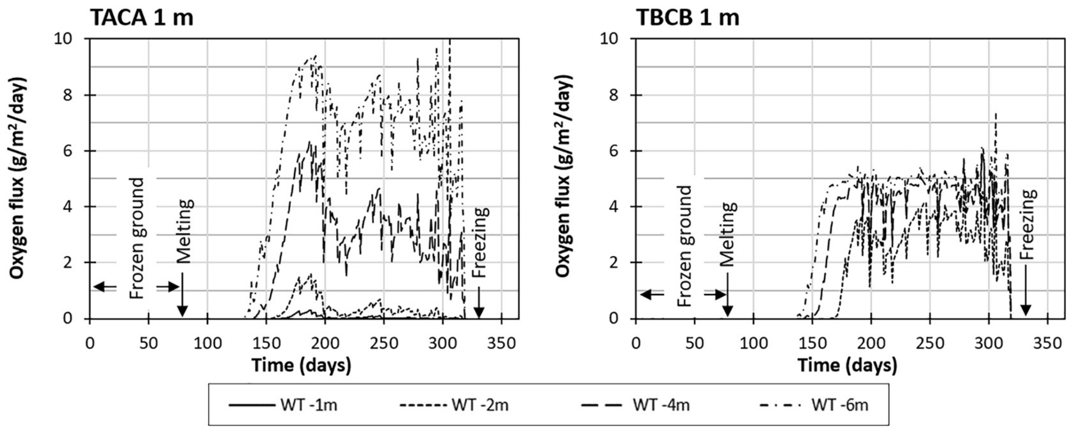

Fig. 5. Simulated daily oxygen fluxes through the 1 m thick monolayer covers CA and CB placed over reactive tailings TA and TB, respectively, for various water table depths.

Table 3
Cumulative annual oxygen fluxes in $g/m^2/y$ through a 1 m thick monolayer cover CA and CB placed over reactive tailings TA and TB, respectively, for various water table depths.

	WT −1 m	WT −2 m	WT −4 m	WT −6 m
TACA 1 m	7.2	62	552	1210
TBCB 1 m	–	424	691	803

The behaviour of a 1 m thick monolayer cover CA or CB, placed over reactive tailings TA (Fig. 6) or TB (Fig. 7), respectively, was assessed. Results (pH, sulfate, iron and zinc concentrations) are presented at z = −1 m.

Reactive transport modelling partially confirms the hydrogeological simulations for tailings TA and cover CA: the deeper the water table, the more AMD is generated. For example, when the water table is located 6 m below the tailings-cover interface, the simulated pH at z = −1 m is as low as 1.8 and sulfate concentrations exceed 4000 ppm after 10 years; when the water table is only 1 m below the interface, the pH is 3.8 and sulfate concentrations are < 80 mg/L (Fig. 6). Zinc concentrations are below 4 mg/L after 10 years when the water table is 1 m deep, around 20 mg/L when it is 2 m deep, and 200 and 350 mg/L when it is 4 m and 6 m deep, respectively. Other dissolved metals and contaminants show similar trends (Pabst, 2011).

The hydrogeochemical simulations also provide additional insight compared to the hydrogeological modelling. For instance, it seems that variations in oxygen fluxes between the different cases for TBCB (Table 3) are not associated with significant changes in the water quality (Fig. 7). The pH remains around 2.5 and sulfate concentrations around 2500 ppm after 10 years, regardless of the water table depth. Also, there is still significant AMD production in the case of TACA when the water table is shallower than 2 m, even though the oxygen flux is < 50 $g/m^2/y$. The pH remains between 3 and 4, and sulfate concentrations still exceed 200 mg/L. This could indicate that the usual target of 50 $g/m^2/y$ for the oxygen flux may not be sufficient (or relevant) in the case of pre-oxidized tailings (at least for the period simulated in the models).

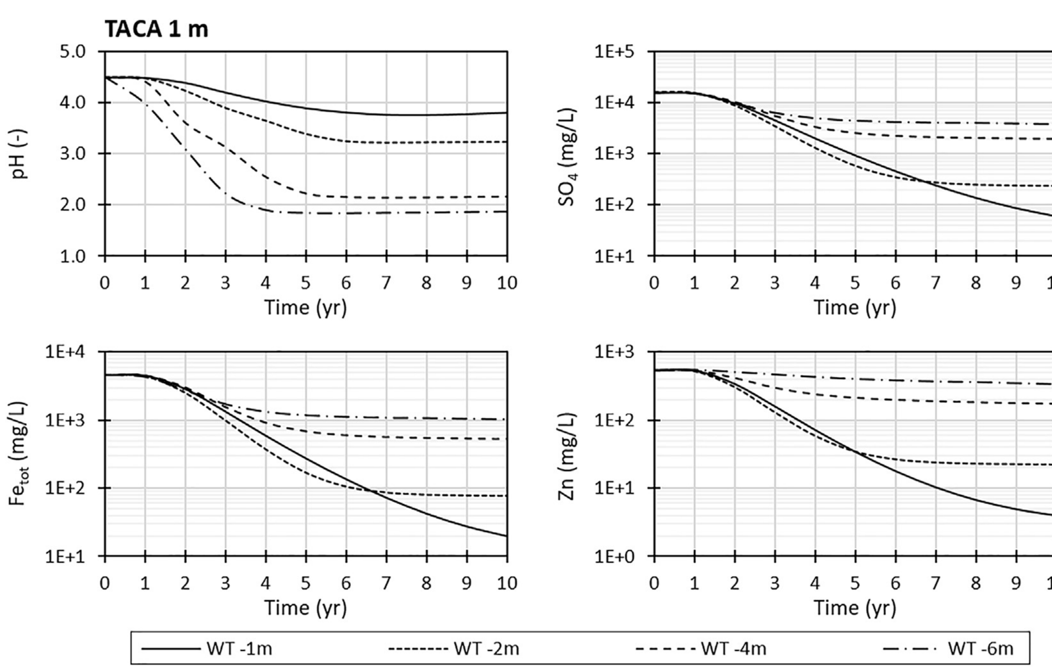

Fig. 6. Evolution of pH and SO_4, Fe_{tot} and Zn concentrations in the leachate water over 10 years for a 1 m thick monolayer cover CA placed over reactive tailings TA. Simulated results obtained with MIN3P are taken 1 m below the reactive tailings-cover interface (z = −1 m), for various water table depths.

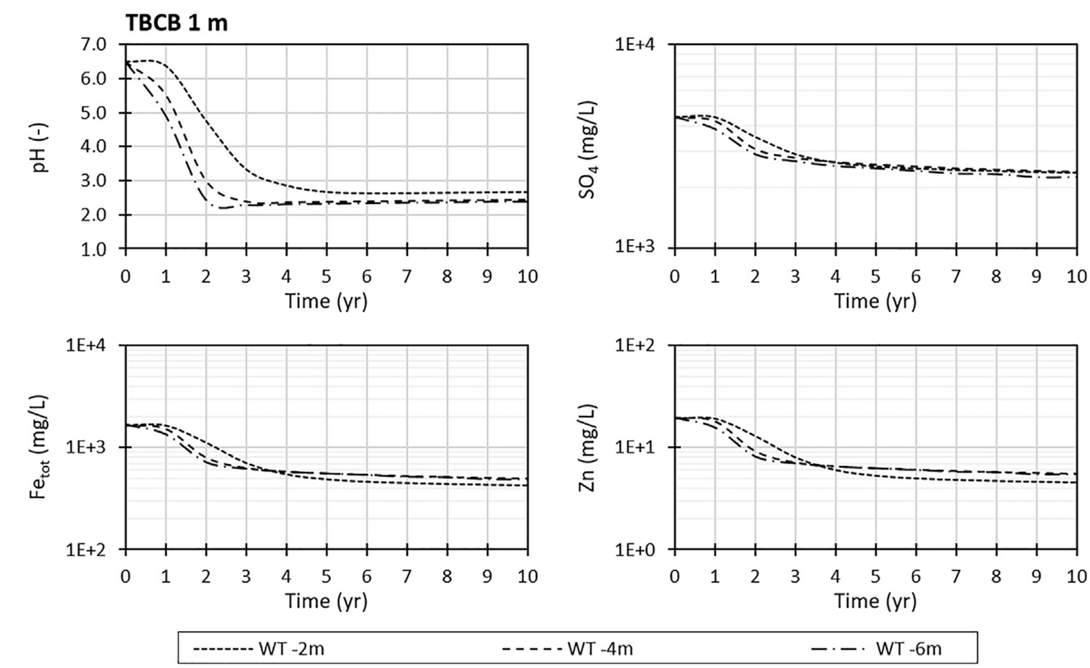

Fig. 7. Evolution of pH and SO_4, Fe_{tot} and Zn concentrations in the water over 10 years for a 1 m thick monolayer cover CB placed over reactive tailings TB. Simulated results obtained with MIN3P are taken 1 m below the reactive tailings-cover interface (z = −1 m), for various water table depths.

4.2. Monolayer covers with different cover thicknesses

4.2.1. Hydrogeological behaviour

The simulations shown above indicated that evaporation was responsible for a large part of water losses in the monolayer covers made of either CA or CB. A thicker cover usually may help reduce adverse effects of evaporation and other atmospheric conditions (like freeze/thaw cycles, for example). Subsequent simulations were thus carried out to assess the effect of cover thickness on its performance for 1 m, 2 m and 4 m thick covers made of non-acid generating tailings CA and till CB. The water table was set 4 m below the tailings-cover interface, a depth at which a 1 m monolayer cover was not sufficient to prevent oxygen diffusion and AMD generation. Variations of the degree of saturation 10 cm above the tailings-cover interface over one year are presented in Fig. 8.

These results indicate that the monolayer cover thickness has a limited effect on the degree of saturation, 10 cm above the tailings-cover interface (Fig. 8): the degree of saturation between the different simulated cases is relatively similar during most of the year, independent of the cover thickness. However, variations of S_r are more progressive for thicker covers, and less dependent on rain events. In general, the increase in degree of saturation is less in thicker covers upon snow melting, but the degree of saturation remains higher during the summer and autumn.

Simulated profiles of the degree of saturation at day 200 (not shown here, but similar to Fig. 4) confirm the previous observations. Evaporation can significantly decrease the degree of saturation close to the surface, but the bottom of thicker covers remains more saturated than thinner covers. As an example, at 50 cm above the interface, considering a water table located at −4 m, the degree of saturation of cover CA is 56% when the monolayer cover is 1 m thick, 68% when it is 2 m thick and 78% when it is 4 m thick. These monolayer covers, independent of their thickness, appear to be ineffective in maintaining the degree of saturation in the cover around or above 85% throughout the year when the water table is deeper than 4 m.

Reactive tailings TA and TB remain close to saturation (> 95%), independent of the cover thickness. However, pore pressures indicate that evaporation can affect the cover and the reactive tailings (in the

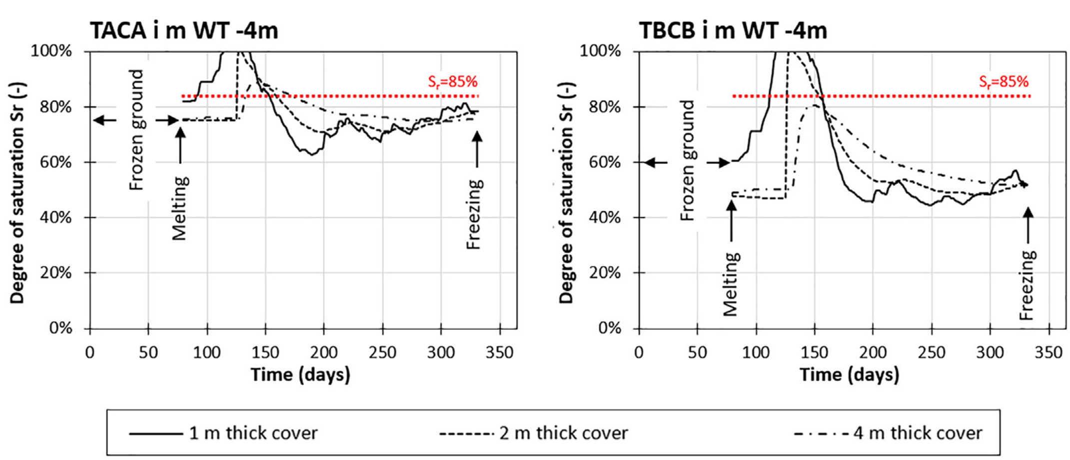

Fig. 8. Evolution of the degree of saturation in monolayer covers (materials CA or CB) of various thicknesses, placed over acid-generating tailings (TA or TB), 10 cm above the surface of the reactive tailings (z = 0.1 m), over one year. The water table is located at z = −4 m.

Table 4
Cumulative annual oxygen fluxes in $g/m^2/y$ through monolayer covers (materials CA or CB) of various thicknesses placed over reactive tailings (TA or TB). The water table is located at $z = -4$ m.

	1 m thick cover	2 m thick cover	4 m thick cover
TACA	552	326	216
TBCB	691	513	365

case of thin covers) over several meters. With a 1 m thick monolayer cover CA, evaporation can affect the reactive tailings up to a depth of several meters, while its effect does not exceed 1 m in the case of a 4 m-thick cover. With CB, the effect of evaporation does not exceed 50 cm below the surface of the cover for all cases.

4.2.2. Oxygen fluxes

The effect of different cover thicknesses on oxygen flux was also assessed (Table 4). Results indicate that a thicker cover is generally more efficient at limiting oxygen ingress. For example, in the case of tailings TA and cover material CA with a water table located 4 m below the tailings-cover interface, the cumulative annual oxygen flux to the tailings is 552 $g/m^2/y$ with a 1 m thick cover and 216 $g/m^2/y$ when it is 4 m thick. Similar results were obtained in the case of tailings TB, where cumulative annual oxygen flux to the tailings decreases when cover thickness increases. In these cases, however, the oxygen flux generally remains above the usual target of 50 $g/m^2/y$, thus confirming previous results, suggesting that such a monolayer cover CA or CB would not be effective for preventing AMD in the case of a relatively deep water table (WT < -2 m).

Previous analyses indicate that keeping the cover CA close to saturation requires a water table no deeper than 2 m below the tailings-cover interface. Increasing the cover thickness can also help decrease the oxygen flux from 62 $g/m^2/y$ (1 m thick cover CA) to 36 $g/m^2/y$ (2 m thick cover) and 28 $g/m^2/y$ (4 m thick cover) (Table 5). However, a two-month drought would significantly decrease the degree of saturation in the monolayer cover CA and the related annual oxygen flux to 179, 100 and 49 $g/m^2/y$ for a 1, 2 and 4 m thick cover, respectively (Table 5). Such a monolayer cover coupled with an elevated water table therefore appears to be highly sensitive to droughts and has a high risk of failing to prevent AMD generation during dry years.

4.3. Bilayer covers

4.3.1. Hydrogeological behaviour

The results obtained for monolayer covers, shown above, indicate that for tailings TA, a single layer cover made of materials CA is susceptible to significant drying when the water table is located > 2 m below the reactive tailings-cover interface (as measured during field surveys). Till CB is even more prone to desaturation. As a result, the oxygen flux is too high, i.e. above the target of 50 $g/m^2/y$.

Alternative solutions have also been considered to improve the water retention within the covers and reduce oxygen flux to the reactive tailings. One option was to create a capillary barrier effect with the addition of an underlying 1 m thick layer made of coarse-grained material (Fig. 3). It can be expected that the tendency of the coarser material to desaturate (because of it low AEV) would produce a capillary barrier effect that limits water loss by downward drainage (e.g. Nicholson et al., 1989; Aubertin et al., 1996, 2009). Two waste rocks (WR1 and WR2), whose properties are based on actual waste rock samples characterized for another project near mine site A (Table 1 and Fig. 1), were considered to create the capillary break.

Table 5
Cumulative annual oxygen fluxes in $g/m^2/y$ through a monolayer cover CA placed over reactive tailings TA, with a water table located at $z = -2$ m, for a normal and a dry year.

TACA	1 m thick cover	2 m thick cover	4 m thick cover
Normal year	62	36	28
Dry year	179	100	49

Simulations were conducted to assess the response of different bilayer cover configurations. In these calculations, the moisture retention layers CA and CB are 1 m and 3 m thick and the underlying coarse layer (made of waste rock WR1 or WR2) is 1 m thick (Fig. 3). The water table is located 4 m below the surface of the reactive tailings (TA or TB).

Results indicate that with a 3 m-thick MRL made of non-acid-generating tailings CA, the degree of saturation 10 cm above the bottom of the MRL (i.e. 10 cm above CA-WR interface) remains above 85% throughout the year; it is somewhat higher (around 90%) when WR2 is used as a capillary break (Fig. 9). However, the degree of saturation in a 1 m thick MRL becomes < 80% after day 160 under the same conditions. In the case of till CB, the degree of saturation 10 cm above the tailings-cover interface is around 50 to 60% for a large part of the year, and cover thickness seems to have little influence on the results. Drying of cover CB is thus more pronounced than with cover CA.

The corresponding degree of saturation profiles (confirmed by suction profiles; not shown here) indicate that a capillary barrier effect develops between the MRL and the coarse-grained material layer. The capillary break seems to be more efficient when using WR2. Additional simulations with WR2 (Pabst, 2011) indicate that there is no significant effect of the water table depth in this case. Nevertheless, evaporation contributes to the desaturation of the cover.

4.3.2. Oxygen fluxes

The oxygen flux was calculated for bilayer covers made of materials CA and CB and using waste rocks WR1 and WR2 (Table 6). The results for the different configurations and water table locations agree with the observations made on the degree of saturation (Fig. 9). In these bilayer covers, there is desaturation of the MRL, resulting in high oxygen fluxes. Only a thick retention layer (3 m) with a capillary break made of waste rock WR2 would keep the oxygen flux below 50 $g/m^2/y$. A bilayer cover made of 1 m of WR2 and a 3 m thick CA MRL give an oxygen flux below 8 $g/m^2/y$.

4.3.3. Water chemistry

The behaviour of bilayer cover systems was also simulated with the code MIN3P. Reactive transport modelling indicates improved water quality after 10 years, 1 m below the tailings surface when the MRL is 3 m thick (Fig. 10). For example, the pH is around 3.8 for both tailings TA and TB while it is closer to 2.1 and 2.5, respectively, for a 1 m thick layer. Sulfate, iron and zinc concentrations are also lower with a 3 m thick MRL, sometimes by orders of magnitude. However, the simulation results also indicate that oxidation and acid mine drainage continues below bilayer covers: pH remains acidic and sulfate concentrations can exceed 1000 ppm even when the oxygen flux is below 50 $g/m^2/y$ (e.g. TB WR2 CB 3 m). A bilayer cover made of a 1 m thick WR2 capillary break and a 3 m thick CA MRL can, however, decrease sulfate and heavy metal concentrations significantly. Nonetheless, further simulations (not shown here) indicate that there would be a significantly decreased performance during dry spells, following a stronger desaturation of the MRL due to evaporation.

4.4. Three-layer covers with capillary barrier effects (CCBE)

4.4.1. Hydrogeological behaviour

The water flow simulations presented above have shown that evaporation is responsible for a large part of the moisture losses and consequent desaturation of the MRL. In such cases, a surface layer made of sufficiently coarse-grained material (e.g. waste rock; Kalonji Kabambi, 2016; Kalonji Kabambi et al., 2017) can be used as an evaporation barrier to prevent water losses to the atmosphere (Barbour and Yanful,

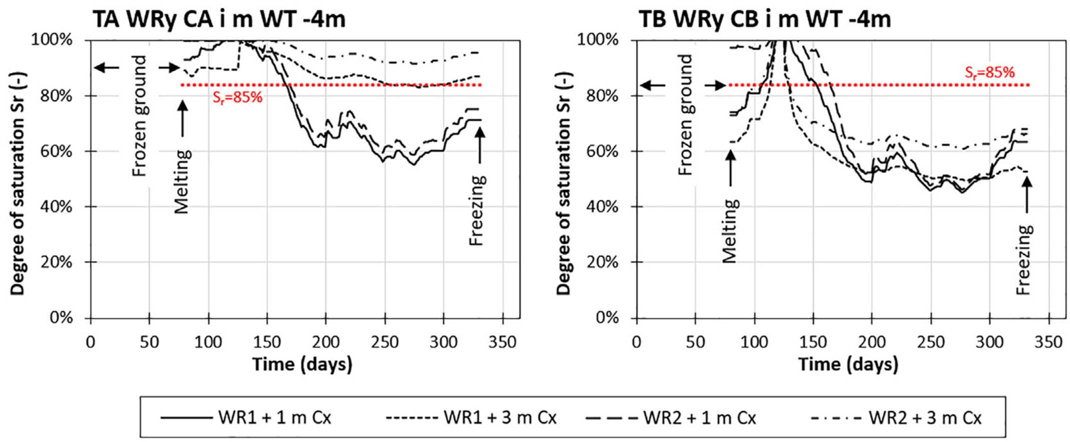

Fig. 9. Evolution of the degree of saturation in bilayer covers (see text for details) placed over acid-generating tailings (TA or TB), 10 cm above the surface of the reactive tailings (z = 0.1 m), over one year. The water table is located at z = −4 m.

Table 6
Cumulative annual oxygen fluxes in $g/m^2/y$ through bilayer covers (see text for details) placed over acid generating tailings (TA or TB). The water table is located 4 m below the tailings-cover interface.

	1 m thick cover	2 m thick cover	3 m thick cover
TA WR1 CA	663	159	78
TA WR2 CA	571	55	6.1
TB WR1 CB	537	369	290
TB WR2 CB	528	200	34

1994; Aubertin et al., 1995, 1999, 2002; Dagenais et al., 2006). Additional simulations were therefore conducted to evaluate this option for three-layer covers consisting of a 1 m-thick waste rock layer (WR1 or WR2) placed on top of reactive tailings (TA or TB), a 1 m thick MRL (CA or CB), and another 0.5 m waste rock layer added on top (Fig. 3). The latter thickness was shown to be sufficient to limit moisture loss by evaporation (for such climatic conditions) in previous studies (Dagenais, 2005; Dagenais et al., 2006). In these simulations, the water table is located 4 m below the tailings-cover interface, which is representative of field conditions.

With such a CCBE, the degree of saturation in the MRL CA remains above 90% at all time (Fig. 11a). The two waste rock layers are desaturated ($\theta = \theta_r$) as indicated by the degree of saturation profile (Fig. 11b). The two coarse layers (capillary breaks) thus create a strong contrast with the CA tailings. The desaturated waste rock layer at the base prevents water from flowing down to the reactive tailings, while the layer on top efficiently limits moisture rise. Waste rock WR2, which is slightly coarser than WR1 (Fig. 1), appears to be more efficient in the role of a capillary break. With till CB, only the capillary break made of WR2 keeps the MRL sufficiently saturated (above 85%). The contrast between CB and WR1 is not pronounced enough and in this case the degree of saturation 10 cm above the bottom of the MRL is around 70% in the summer and autumn (well below the target of 85%).

The effect of a dry year (Fig. 2) was also investigated with additional simulations. Results (dashed lines in Fig. 11a,b) show that the absence of precipitation during two months in the summer would only slightly affect the MRL. The degree of saturation in the retention layer

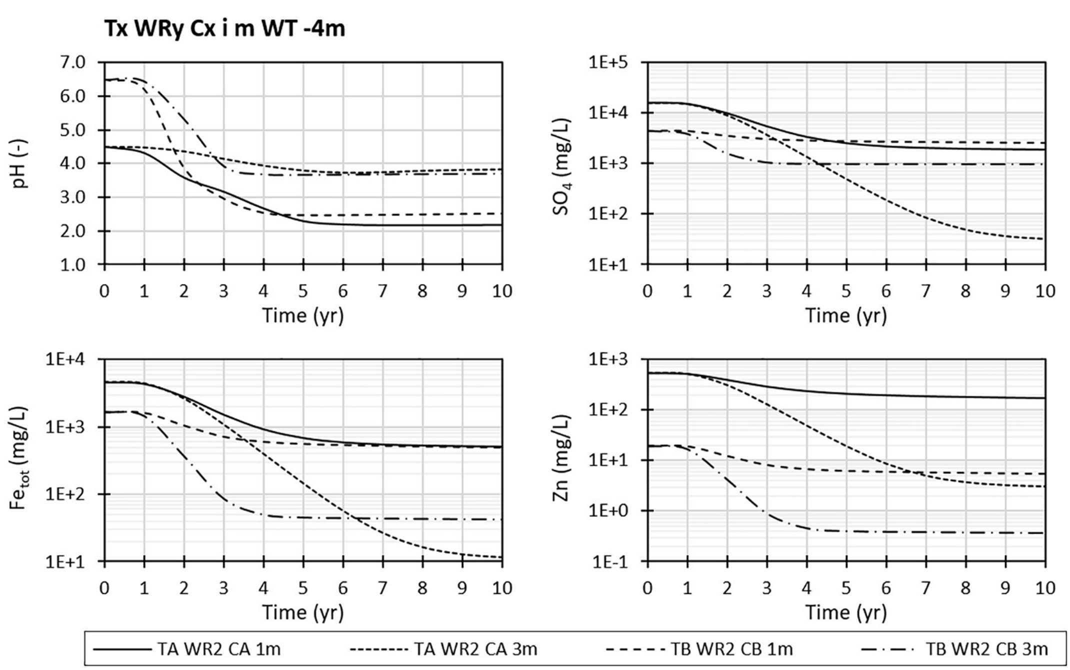

Fig. 10. Evolution of pH and SO_4, Fe_{tot} and Zn concentrations in the water over 10 years for bilayer covers placed over reactive tailings (TA or TB). Simulated results obtained with MIN3P are taken 1 m below the reactive tailings-cover interface (z = −1 m). The water table is located at z = −4 m.

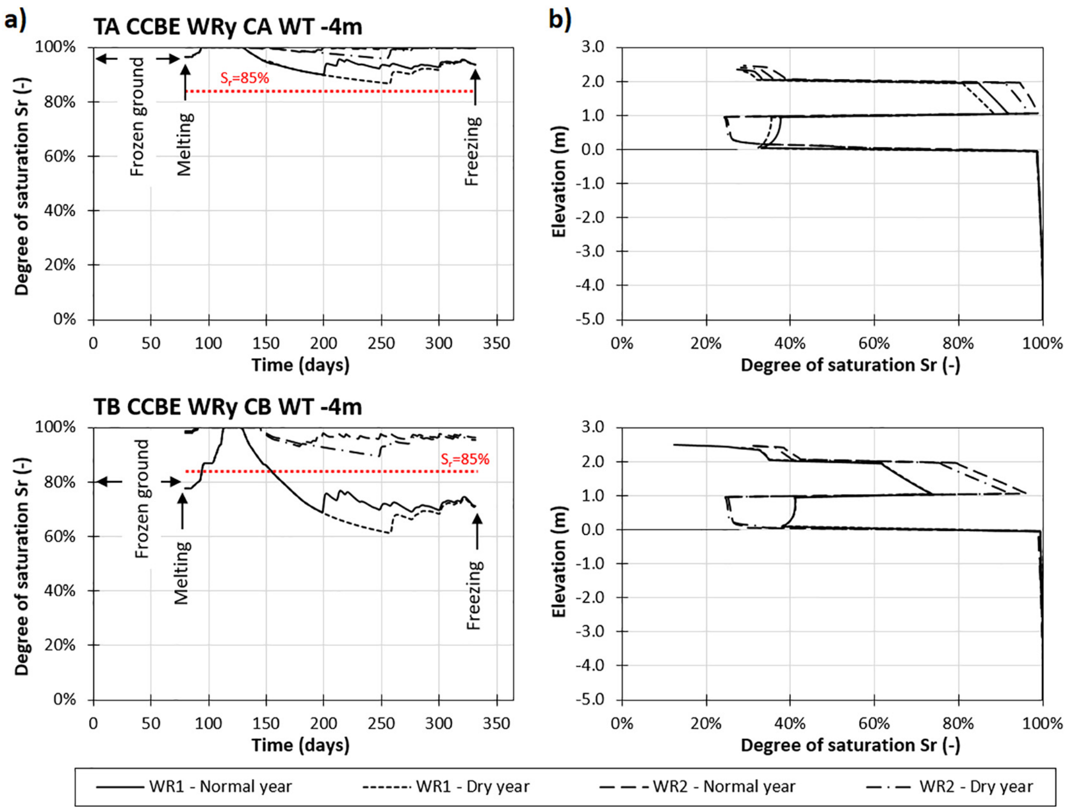

Fig. 11. Hydrogeological behaviour of CCBEs (see text for details) placed over acid generating tailings (TA or TB). The water table is located at z = = − 4 m. (a) Evolution of the degree of saturation in the MRL, 10 cm above the bottom of the layer (z = 1.1 m) over one year; (b) vertical profile of the degree of saturation at day 200 (mid-summer).

would remain high (above 95%) with waste rock WR2 used as a capillary break.

4.4.2. Oxygen fluxes

With a three-layer CCBE, the cumulative annual oxygen flux through the cover (Table 7) is much less than in the previous simulated cases; it is generally well below 50 g/m^2/y, except for the case with waste rock WR1 and till CB. In all cases, the oxygen flux with CB is at least one order of magnitude greater than with CA, which confirms that till CB is less suitable as a MRL than the non-acid-generating tailings CA. Results also indicate that the difference in fluxes for covers using WR1 or WR2 can be greater than two orders of magnitude. The configuration CA and WR2 gives the best results with a simulated oxygen flux < 0.1 g/m^2/y.

4.4.3. Water chemistry

Reactive transport modelling was carried out to assess the effect of well-designed three-layer CCBEs on the water quality. Only cases where oxygen fluxes are below the target of 50 g/m^2/y are shown here. The results shown in Fig. 12 indicate significant differences between the various configurations, despite oxygen fluxes being low and below 50 g/m^2/y. Only the case where F_{O2} < 0.1 g/m^2/y (TACAWR2) allows for a pH around neutrality with a pH = 6 (which is close to that of infiltrating water in the model). Sulfate and iron concentrations are < 1 mg/L and zinc concentrations are around 0.01 mg/L. Using WR1 as a capillary break (instead of WR2) increases the oxygen flux by more than two orders of magnitude, but it remains relatively low (around 10 g/m^2/y). In the latter case, the pH after 10 years (1 m below the tailings-cover interface) remains below 4, while the sulfate and iron concentrations are around 10 mg/L and the zinc concentrations exceed 2 mg/L. The case where CB is used with WR2 represents an intermediate case in which the water pH 1 m below the tailings-cover interface is around 5.0; iron and zinc concentrations are low, but sulfate concentrations remain above 400 mg/L. This suggests that oxidation of tailings TB is continuing. The case TBWR1 was not simulated due to the very high oxygen fluxes (> 200 g/m^2/y, Table 7). The simulations also show that water quality (1 m below the tailings-cover interface) tends to stabilize after around 4 years. This is in accordance with observations made, for example, at the Lorraine mine site by Dagenais (2005) and Bussière et al. (2009).

Table 7
Cumulative annual oxygen fluxes in g/m^2/y through CCBEs (see text for details) placed over acid generating tailings (TA or TB) for a normal and a dry year. The water table is located 4 m below the tailings-cover interface.

	Normal year	Dry year
TA CEBC WR1 CA	11	26
TA CEBC WR2 CA	0.03	0.34
TB CEBC WR1 CA	241	297
TB CEBC WR2 CA	2.1	7.6

5. Discussion

The numerical results shown here, which include data supported by experimental observations (Pabst, 2011; Pabst et al., 2014, 2017a,b), indicate that a monolayer cover may not be efficient enough to prevent AMD generation for the conditions considered here, particularly where the water table is deeper than 2 m below the surface of the reactive tailings. For reactive tailings TA, monolayer covers CA became desaturated because of the combined effects of both drainage and evaporation. Reactive tailings are thus strongly affected by evaporation, and even though they remain close to saturation, their degree of saturation is lower than expected.

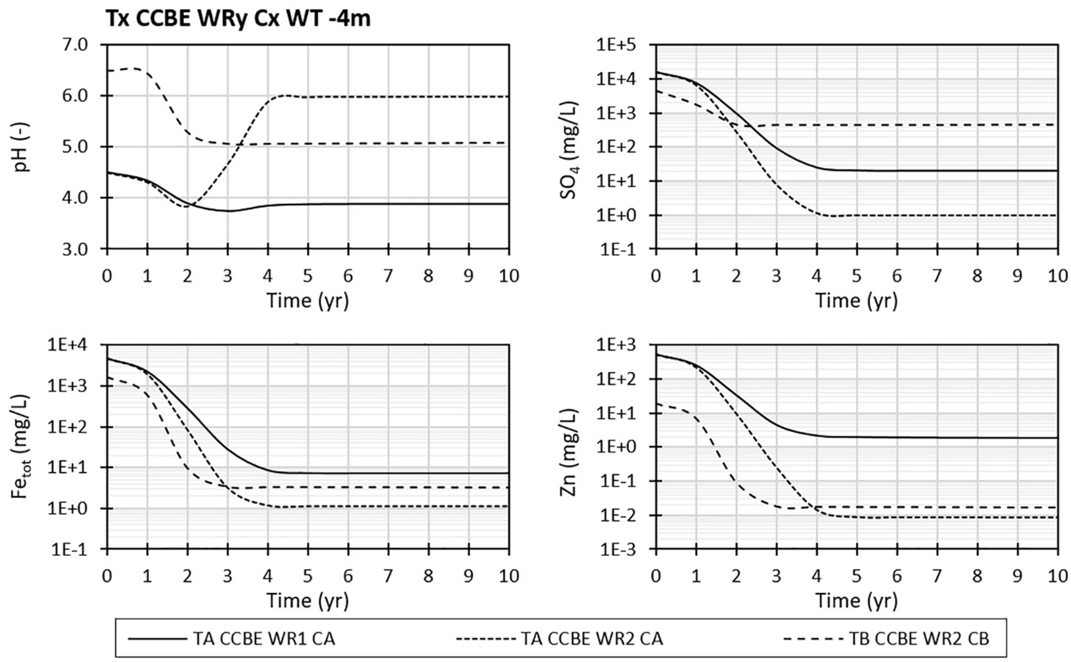

Fig. 12. Evolution of pH and SO_4, Fe_{tot} and Zn concentrations in the water over 10 years for CCBEs (see text for details). Simulated results obtained with MIN3P are taken 1 m below the reactive tailings-cover interface ($z = -1$ m). The water table is located at $z = -4$ m.

For reactive tailings TB, material CB is a somewhat better evaporation barrier than material CA, due to its relatively low AEV, but coarse-grained tailings TB desaturate mainly because of downward drainage. Overall however, both monolayer covers CA and CB appear to be too fine to efficiently play the role of an evaporation barrier and too coarse to develop an efficient capillary barrier effect with the underlying tailings TA and TB. Simulations of water distribution and oxygen fluxes (combined with reactive transport modelling) indicate that under such conditions, oxygen would migrate relatively quickly to the reactive tailings, and thus contribute to the generation of AMD.

The contribution of evaporation to cover desaturation was further assessed by simulating bilayer covers. In this case, the top MRL does not remain close to saturation, despite the presence of a strong capillary break formed by the underlying waste rocks WR1 and WR2, which prevents water losses by downward drainage. Three-layer CCBEs were also simulated, based on recommended designs for such covers (Aubertin et al., 1994, 1995, 1999, 2002; Ricard et al., 1997, 1999; Nastev and Aubertin, 2000; Dagenais et al., 2006), i.e. a 1 m-thick waste rock layer (WR1 or WR2) placed between the reactive tailings and the MRL (1 m thick finer grained material, CA or CB), with another 0.5 m waste rock layer on top. The simulation results clearly show that such a cover would better reduce the oxygen flux to the tailings, even during fairly long drought periods. These simulations indicate, however, that even with this configuration, the till CB is not efficient enough as an MRL material (with the waste rock WR1) because of its weak water retention characteristics.

This extensive numerical study also shows some of the advantages of combining hydrogeological and geochemical models to assess the performance of cover systems, particularly when the tailings are already oxidized. In this case, relying only on typical targets for oxygen fluxes may be misleading. Even though the oxygen flux remains below 50 g/m^2/y, which is a practical objective for cover design (e.g. Aubertin et al., 1999; Dagenais, 2005), hydrogeochemical simulations indicate that significant AMD generation may continue over time due to the pre-oxidation state of the tailings and the already acidic and contaminated pore waters that promote indirect oxidation (e.g. Nicholson, 1994; Williamson and Rimstidt, 1994; Kirby et al., 1999; Nordstrom, 2000). These new results also indicate that recommendations made by Ouangrawa et al. (2009, 2010) following previous investigations on fresh/non-oxidized tailings may not be appropriate in this case; the design criterion based on maintaining the water table at a depth below one-half of the AEV of the reactive (fresh) tailings (approximately 3 to 4 m and 1 m for TA and TB, respectively) would not be directly applicable to already oxidized tailings. In such cases, it would be preferable to elevate the water table closer to the surface of the reactive tailings (as demonstrated also by the column tests conducted by Bussière et al., 2011). An oxygen flux of around 1 g/m^2/y also seems a better target to prevent further AMD generation in the particular case of sites A and B (Table 7 and Fig. 12). Therefore, even in this case, the simulations indicate that it could take several years before actually observing an improvement of the leachate quality (which is not unusual for the reclamation of old mining sites; e.g. Dagenais, 2005; Bussière et al., 2009). Also, geochemical simulation results were assessed 1 m below the tailings surface and it could take even longer before improvements are observed at the outflow.

The various simulated scenarios also confirmed the importance of carefully choosing the cover material(s). For the capillary break material, waste rock WR2 appears significantly more efficient to prevent desaturation of the MRL and lower the oxygen flux through the 3-layer CCBE. Using a coarser waste rock may, however, increase the risk of developing localized flows under highly saturated conditions (e.g. Dawood and Aubertin, 2014; Broda et al., 2015; Lahmira et al., 2016); these effects are not considered in the numerical models applied here. Using a very coarse waste rock as a protection and evaporation layer may also induce local advective fluxes that could increase evaporation from the MRL. It is therefore recommended to further investigate the optimal grain size before using waste rocks in layered cover systems (Kalonji Kabambi, 2016; Kalonji Kabambi et al., 2017).

Much effort was devoted in this investigation to calibration of the models and exhaustive parametric analyses, but some uncertainties nonetheless remain. For example, it is challenging and relatively uncertain to extrapolate laboratory results to field conditions, especially when considering in situ heterogeneity and 2D or 3D effects (particularly close to the boundaries of the tailings impoundments). For instance, tailings TA characterized here are relatively fine-grained compared to some of the tailings sampled elsewhere on site A (e.g. Gosselin, 2007; Pabst, 2011) and to typical hard rock tailings from other sites (e.g. Aubertin et al., 2002; Bussière, 2007); thus, tailings TA are only partly representative of the conditions on the site. Additional tests and simulations would be required to assess the responses for different grain-sizes (and compositions) in situ. Also, even though column tests were conducted over almost two years, the required lifetime of the cover on such type of tailings impoundment could exceed hundreds of years (e.g. Vick, 2001; Aubertin et al., 2002, 2016). Additional uncertainties are therefore related to the very long-term evolution of physical properties and mineralogical and chemical composition of the materials, as well as reaction kinetics. Also, the indirect oxidation reactions were taken into account here through a calibration of the reactive transport models based on column tests (Pabst et al., 2017b); further work is recommended to improve reactive transport simulations, especially in terms of reaction kinetics and pH dependency. Finally, it was assumed in the simulations that the water table was constant throughout the year and independent of the cover systems. In practice, it may fluctuate seasonally, and could vary depending on the type of cover. Larger 3D models of the site should therefore be used to assess the effect of varying water table position considering its importance in terms of cover performance (Broda et al., 2014).

Several simplifications were also made when simulating climatic conditions, mainly due to numerical constraints and lack of more accurate climatic data for the two abandoned mining sites. Given the importance of evaporation on the efficiency of a cover system, more emphasis should therefore be placed on this aspect (see for example Wilson et al., 1997). Additional effort should also be devoted to address the effects of climate change, which can be expected to affect the performance of cover systems at limiting oxygen ingress over the long-term (e.g. MEND, 2011; Reinecke and Brodie, 2012).

6. Conclusion

Coupled hydrogeological and geochemical numerical simulations, calibrated and validated on instrumented column tests, were carried out to investigate the performance of different cover systems to reclaim pre-oxidized sulfidic tailings at two impoundments. The degree of saturation, pore water pressure, oxygen flux and leachate quality were considered to assess cover efficiency to limit AMD generation. The effect of various factors, such as water table position, material properties and cover layer thickness were investigated. Alternative reclamation options, including monolayer covers and two- or three-layer covers with capillary barrier effect(s) were also simulated.

The simulations illustrated how the behaviour and efficiency of a monolayer cover placed over reactive tailings depend on specific factors such as its hydraulic properties, its thickness, and more importantly, the water table position. The covers at the two sites are prone to desaturation, especially when the water table is deep below the tailings-cover interface. Under the tested conditions, the simulations tend to indicate that a monolayer cover would not be able to prevent oxygen ingress, except possibly with a highly-elevated water table or a very thick cover. Oxygen flux estimates and reactive transport modelling corroborated this conclusion. Simulation results also tended to indicate that under a given set of conditions, bilayer covers that include a base capillary break layer would not significantly improve their performance (compared to a monolayer cover) due to water losses by evaporation. Three-layer covers with capillary barrier effects (CCBE) appear much more efficient in reducing the oxygen flux and AMD generation. The results further indicate that a well-designed CCBE could maintain its efficiency during relatively long dry spells with minimal water replenishment (under the conditions considered here).

The simulations have also shown that choosing the proper materials for the moisture retaining layer and the capillary break is critical. A rigorous program involving extensive characterization, laboratory experiments and numerical simulations is recommended to properly design an efficient and long-lasting cover system. It is also important to take into account the effects of pre-oxidation of the tailings and the initial porewater chemistry, which may also affect cover performance. This study has further highlighted how coupled hydrogeological and geochemical simulations can lead to important new insights and more realistic results, especially when calibrated on representative experimental tests.

This project, coupled with additional laboratory and field tests (Pabst, 2011; Bussière et al., 2011) and large-scale numerical simulations (Broda et al., 2014; Ethier et al., 2018), contributed to the final design of the cover system for several portions of site A. Additional field work is ongoing to assess in situ cover performance and to confirm laboratory and numerical results (e.g. Ethier et al., 2014, 2017).

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